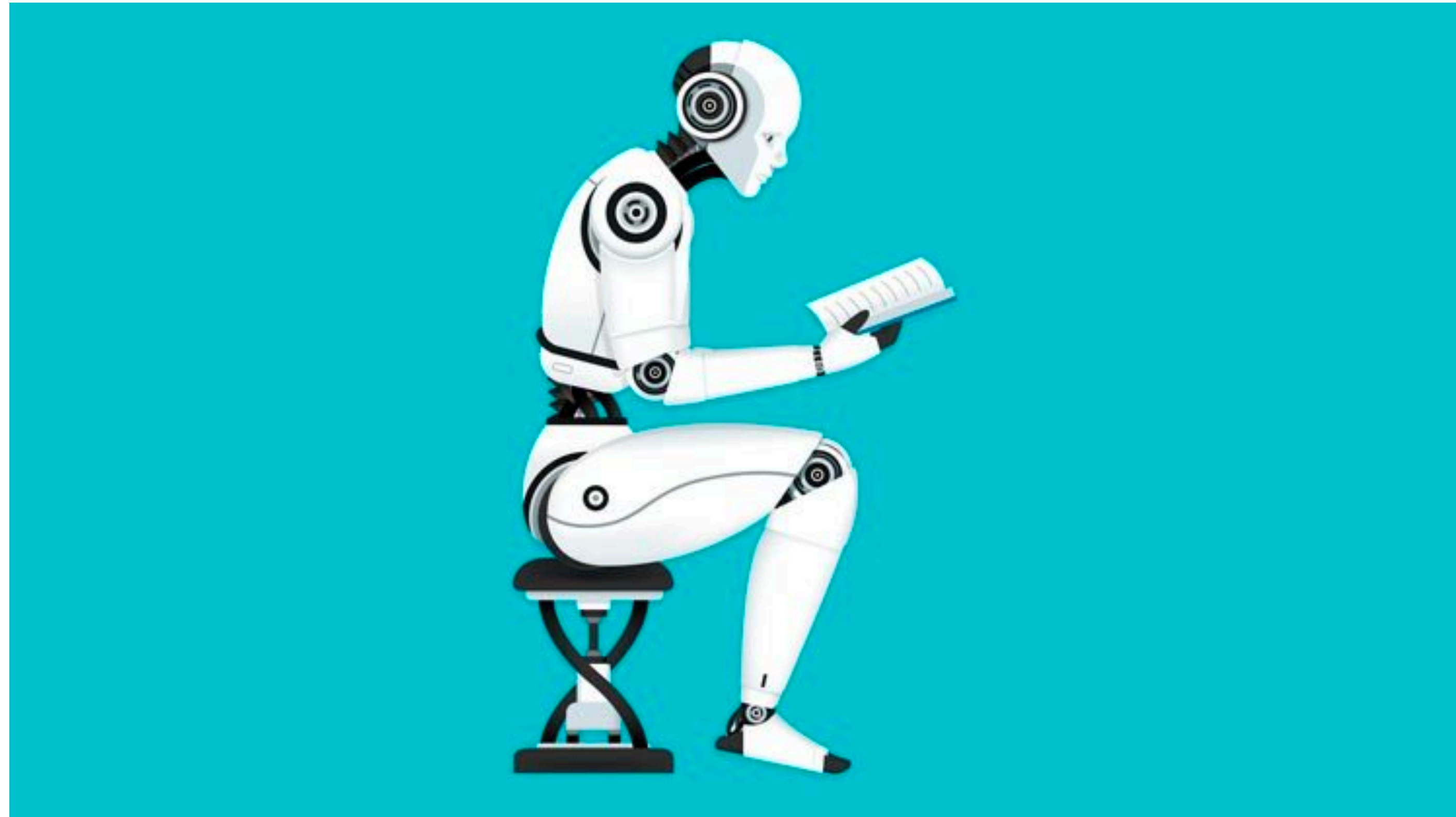


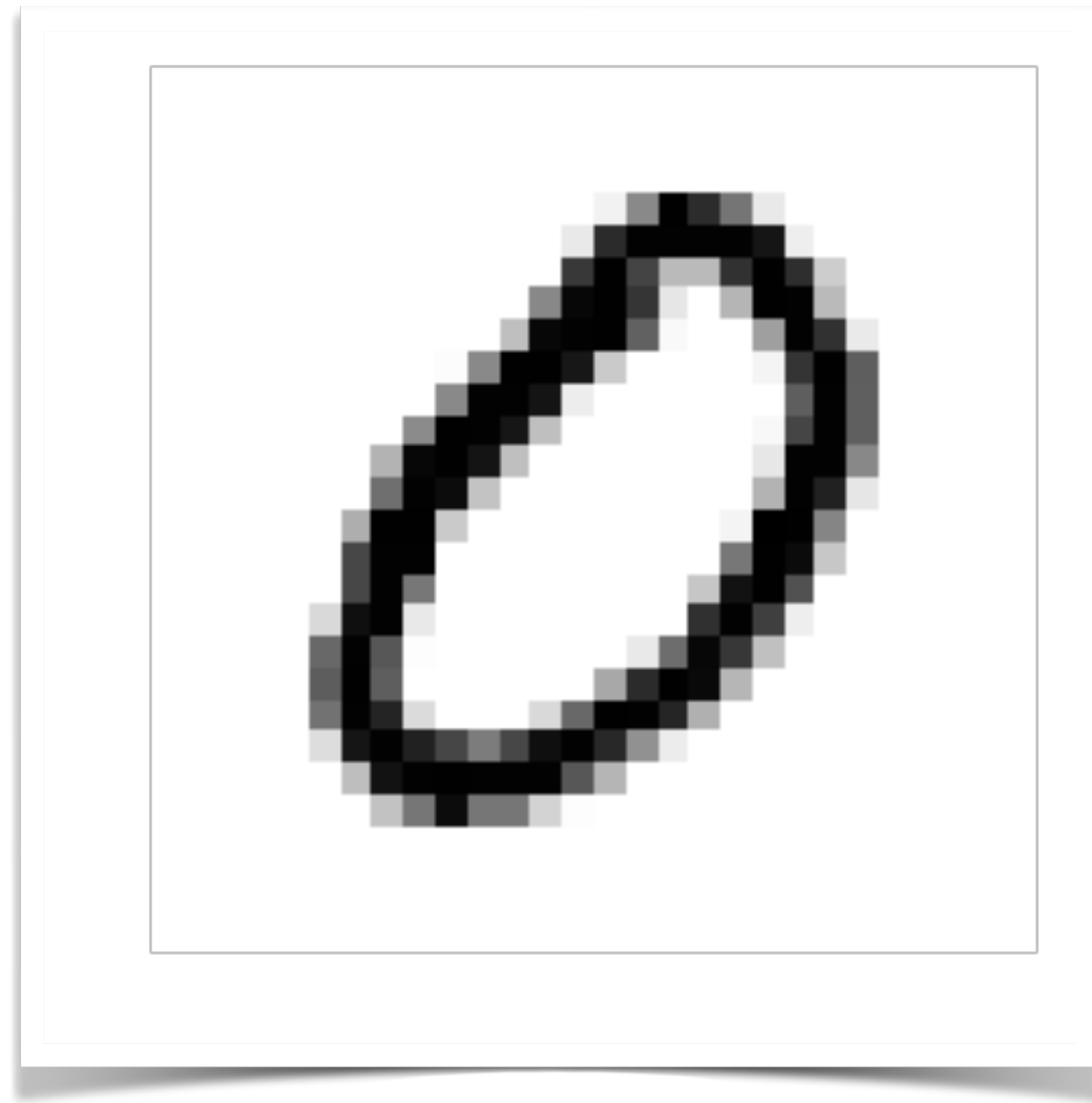


Neural Networks II

Machine Learning | Enginyeria Informàtica

Santi Seguí | 2026-2027

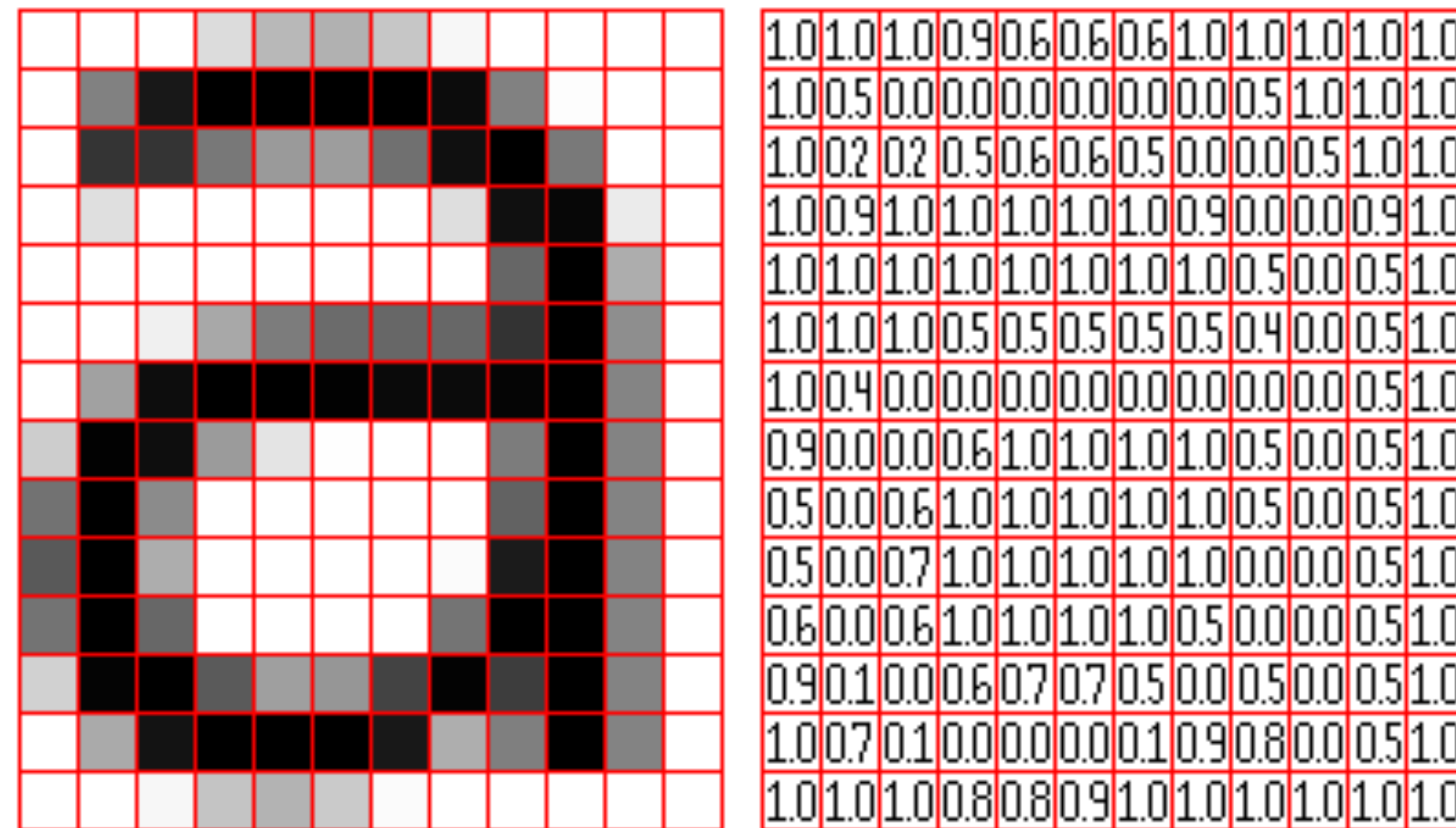
wrap up!



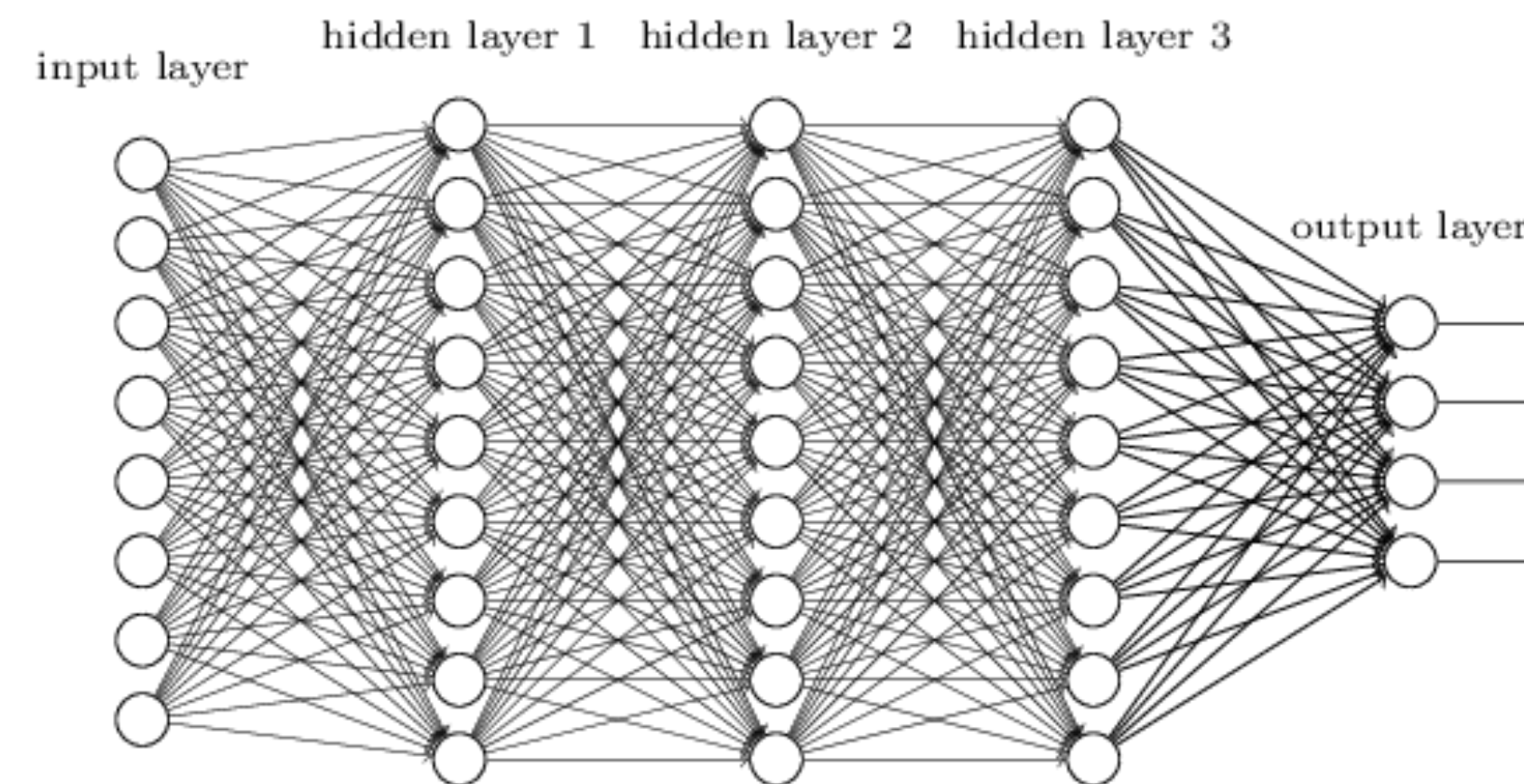
How can we create a Neural Network able to detect the image digit?

Neural Networks for Images?

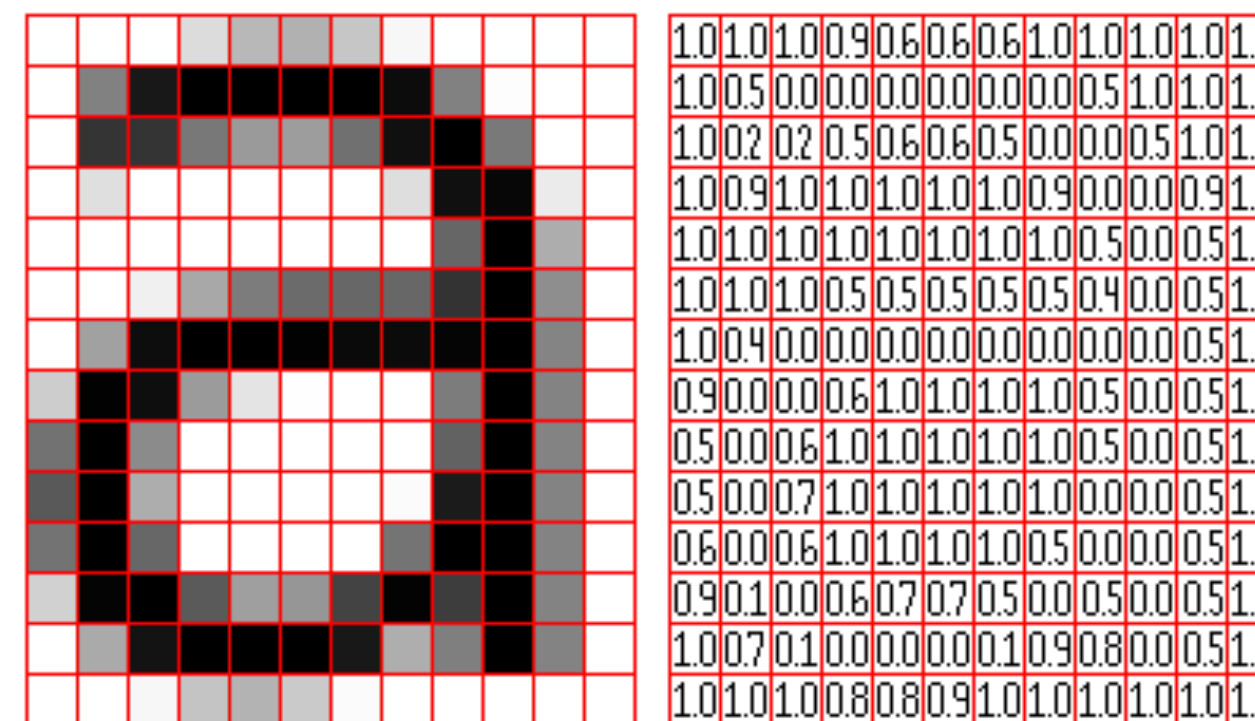
An image is a matrix of size $m \times n \times c$ pixels



Neural Networks for Images?



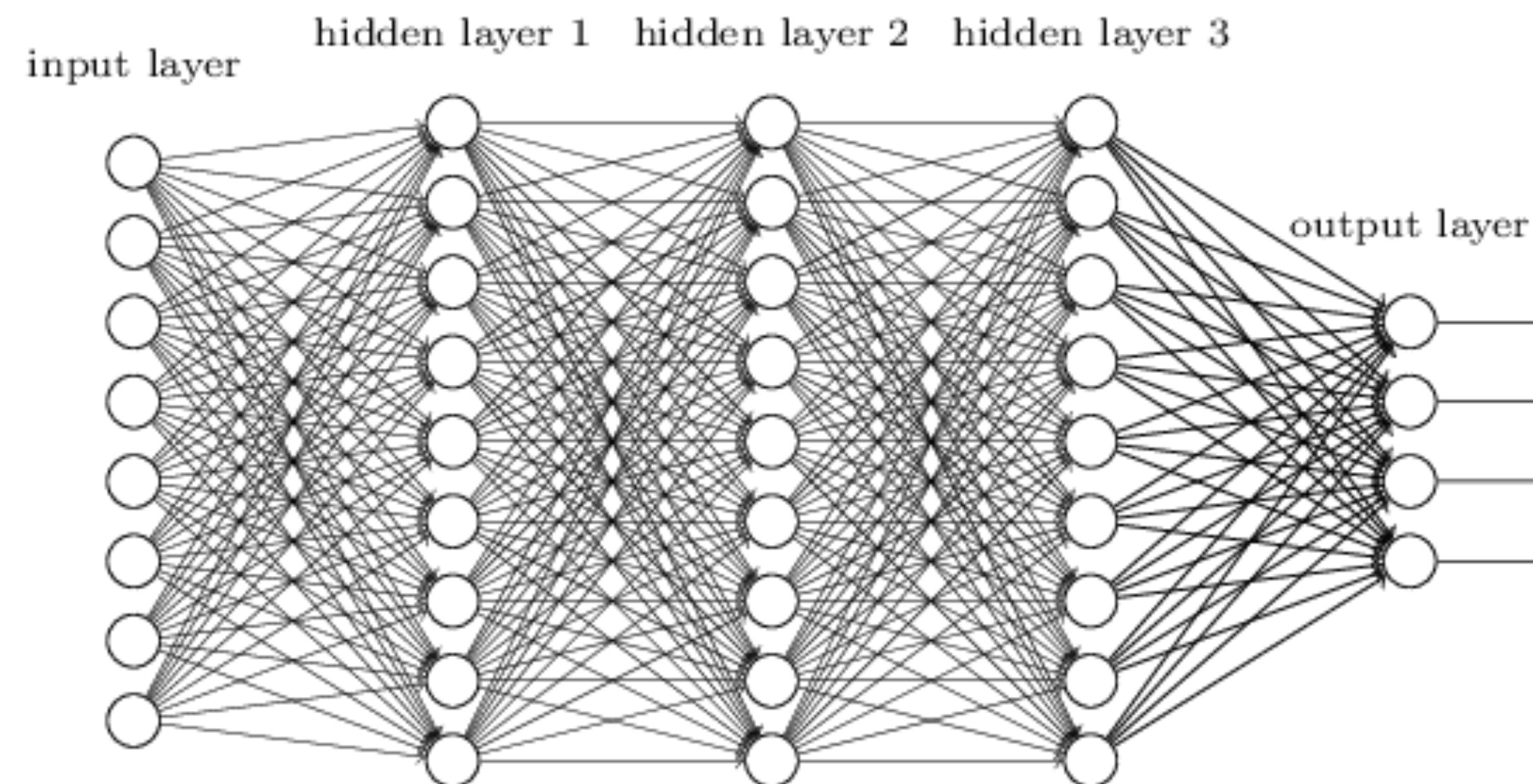
What happens if your data is an image like this one?



Neural Networks for Images?



Neural Networks for Images?



How many parameter does this MLP has?

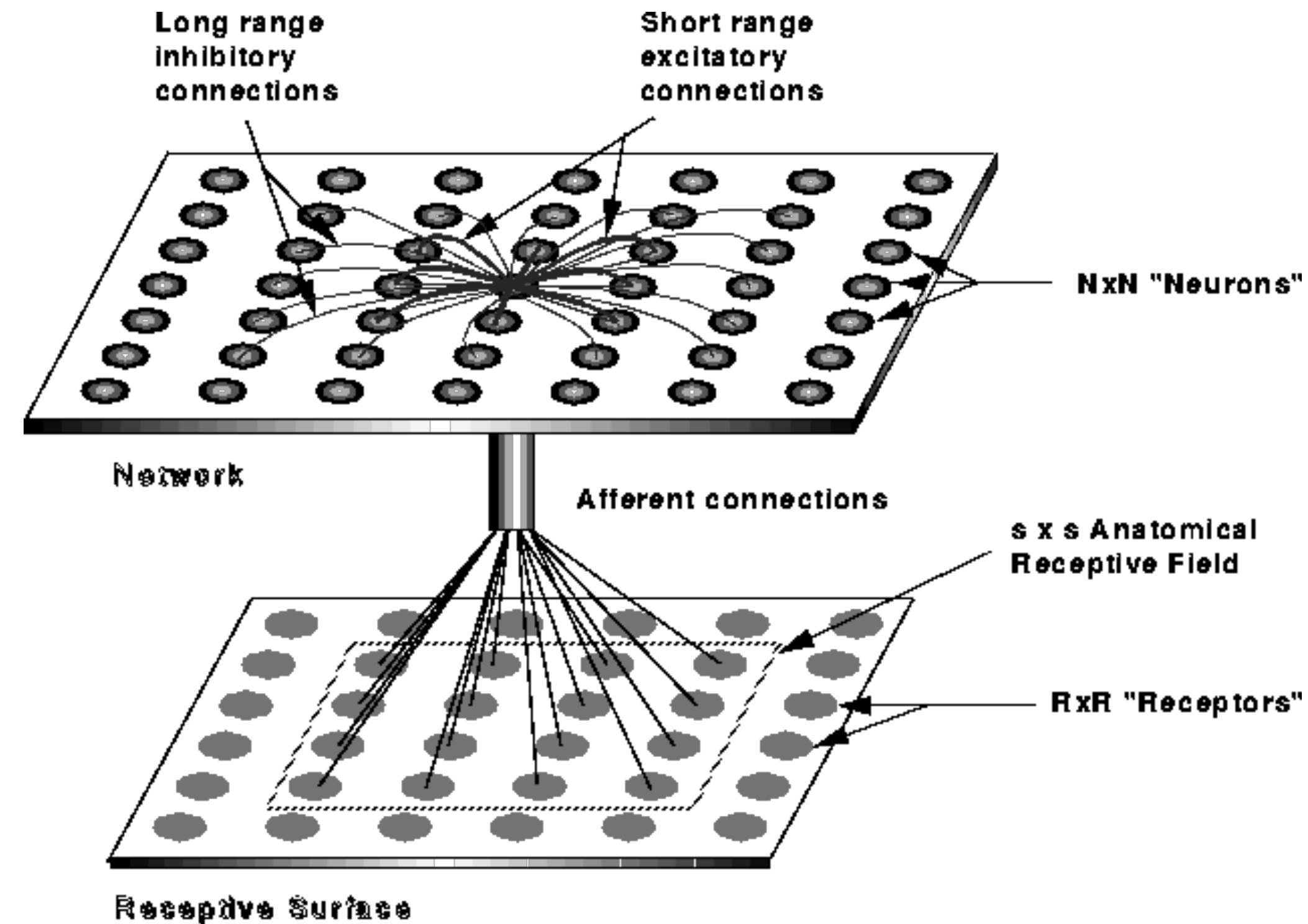
Neural Networks for Images?



Neural Networks for Images

- Input is a standard vector of size $N \times M \times C$
- Imagine a medium resolution color image of 256x256 pixels
- If we think on a Multi Layer Perceptron with just one hidden layer of 256 neurons + an output layer of 1 neuron it will have more than **48 milion** parameters.
- **Does it make sense? Can we do it better?**

Local Receptive Fields



David Hunter Hubel and Torsten Nils Wiesel, 1968

But, in an image:
A dog can appear **anywhere** in the image!



Doesn't matter where they appear,
they look similar anywhere!

Convolutional Neural Networks (CNNs)

- Three main ideas:
 1. **local receptive fields,**
 2. **shared weights,**
 3. **sub-sampling**

Convolutional Neural Networks (CNNs)

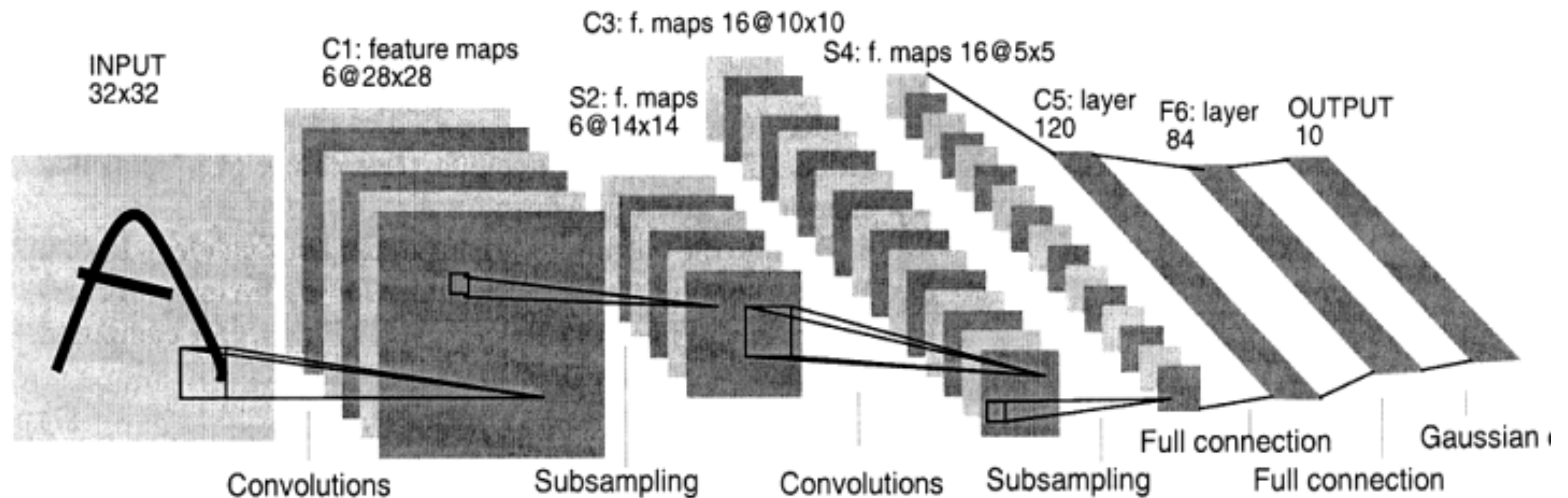
- Repetitive blocks of neurons that are applied across space (for images) or time (for audio signals etc).
- For images, these blocks of neurons can be interpreted as 2D convolutional kernels, repeatedly applied over each patch of the image.
- For speech, they can be seen as the 1D convolutional kernels applied across time-windows.
- At training time, the weights for these repeated blocks are 'shared', i.e. the weight gradients learned over various image patches are averaged.

What is an image convolution?



Input

“Nothing New”

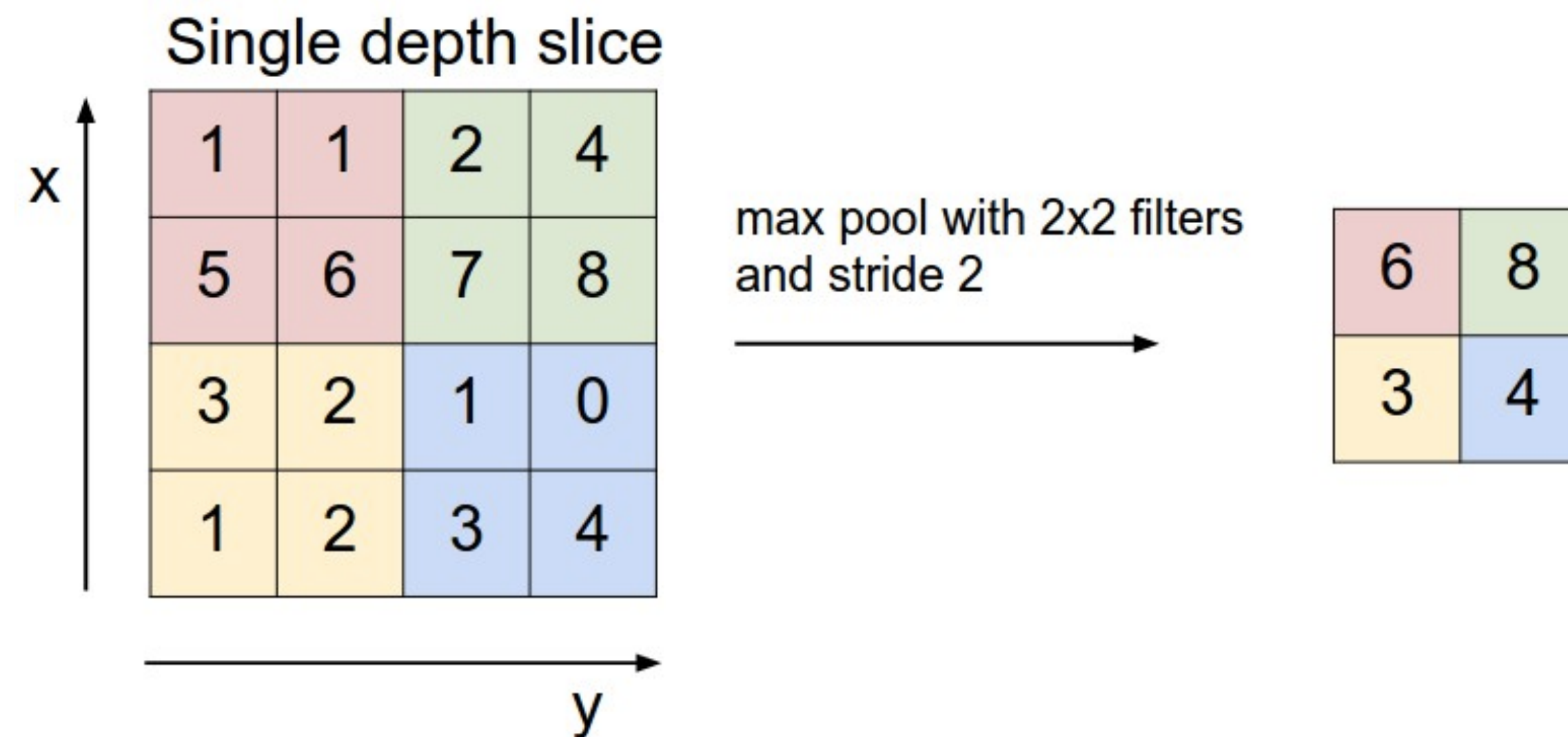


LeCun et al. 1998

Max pooling

Pooling is a way of sub-sampling, i.e. reducing the dimension of the input (or at some hidden layer).

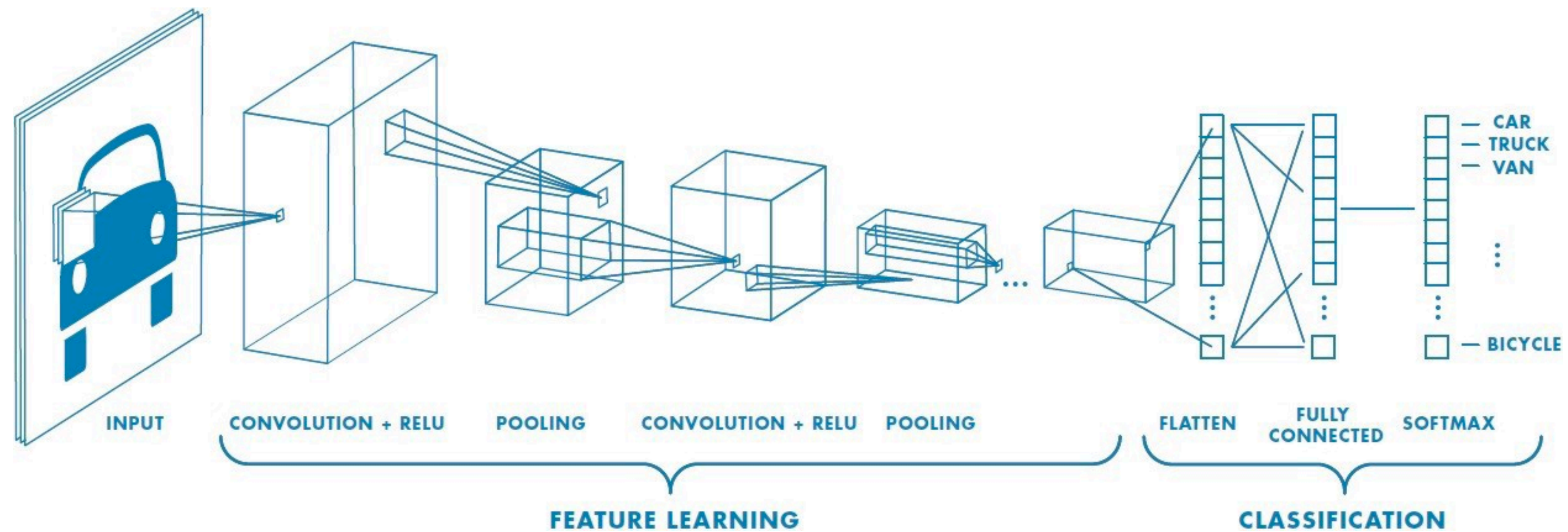
It is usually done after some of the convolutional layers



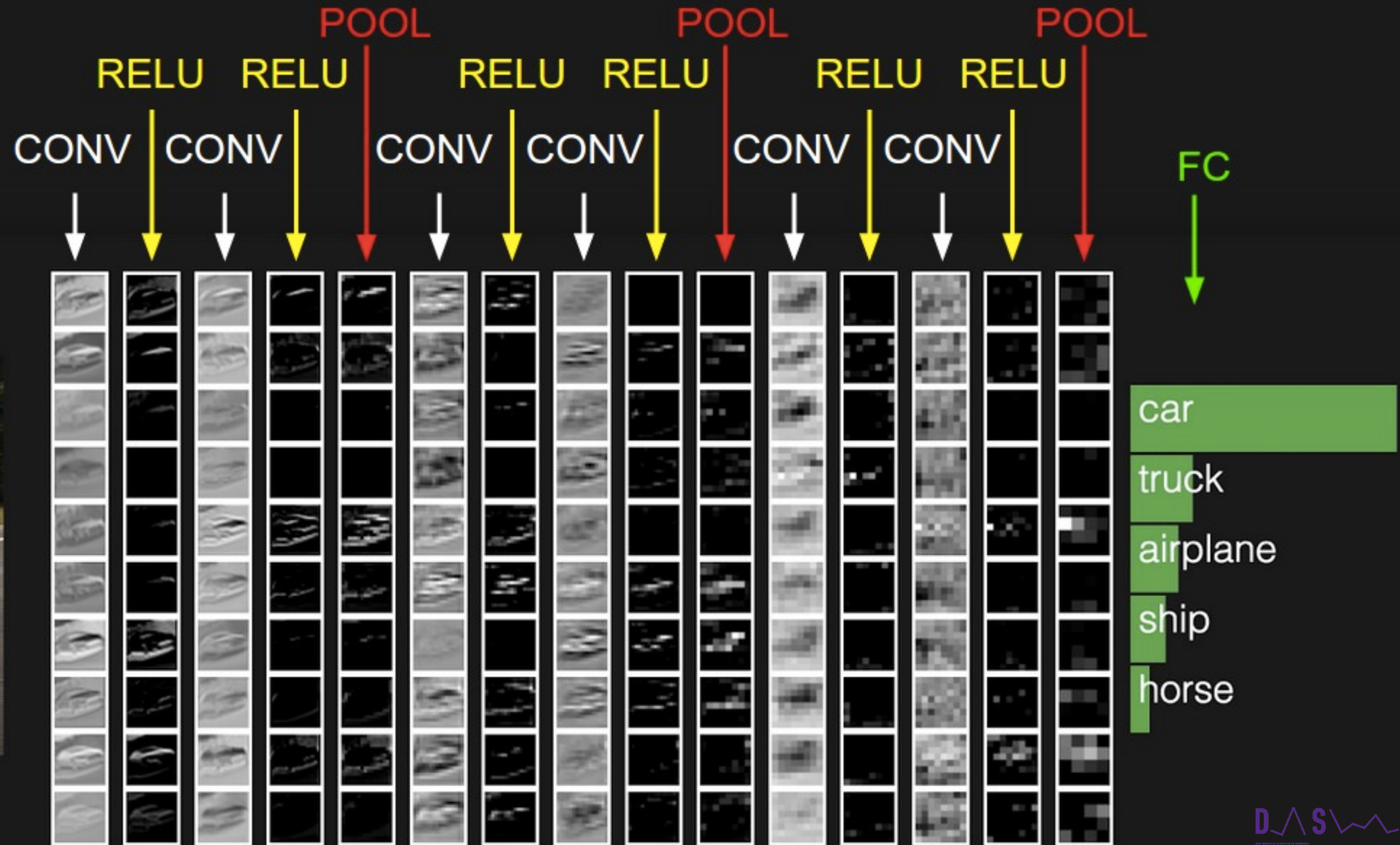
But it is also useful since it provides a form of translation **invariance**

Question:

**Why MAX Pooling
and not
Average Pooling?**

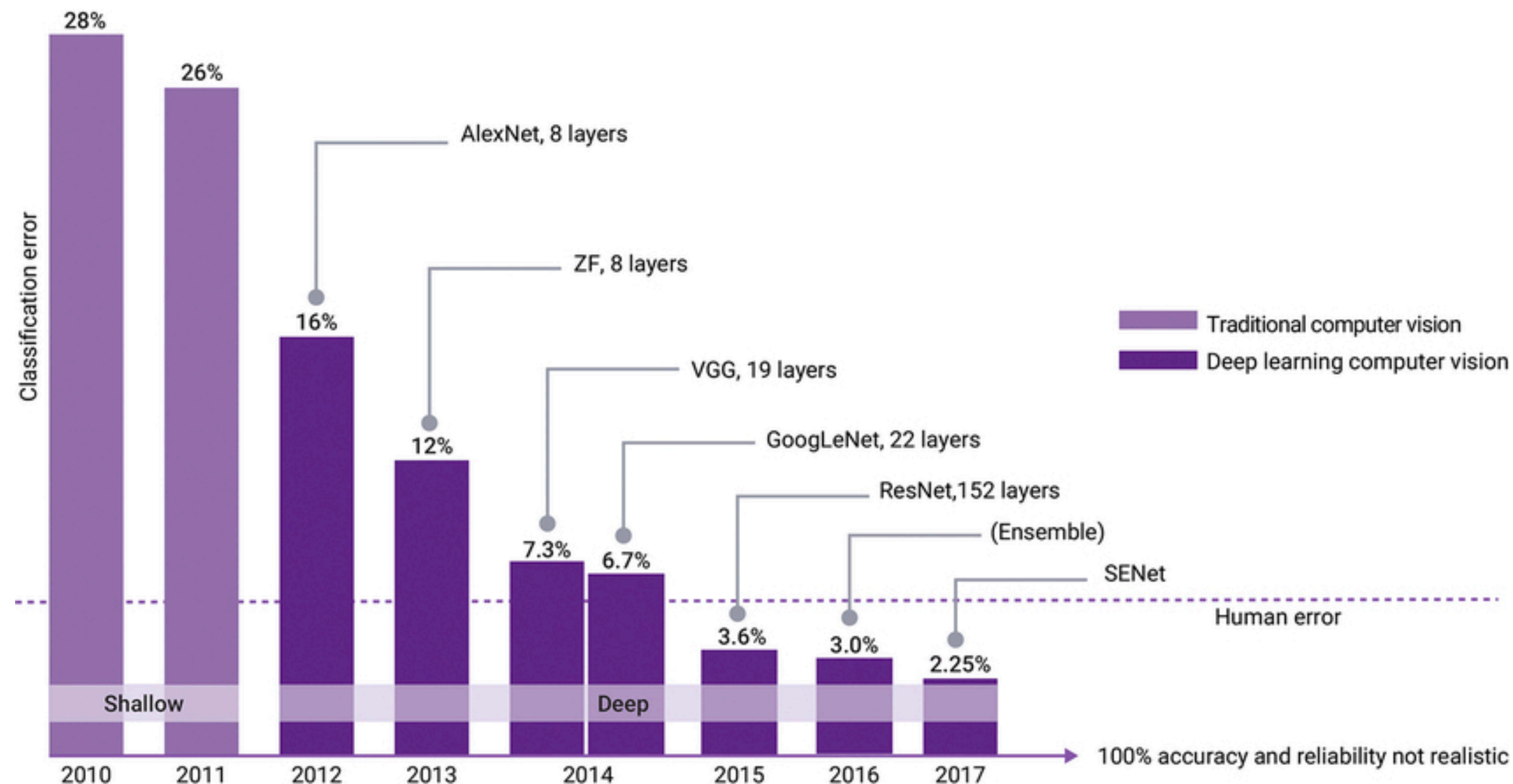


Finally..



Convolutional Neural Networks (CNNs)

In computer Vision the breakthrough resulted in 2011 when Ciresan et.al introduced an algorithm to train these networks by using graphical cards (GPUs)



AlexNet

Bigger model:

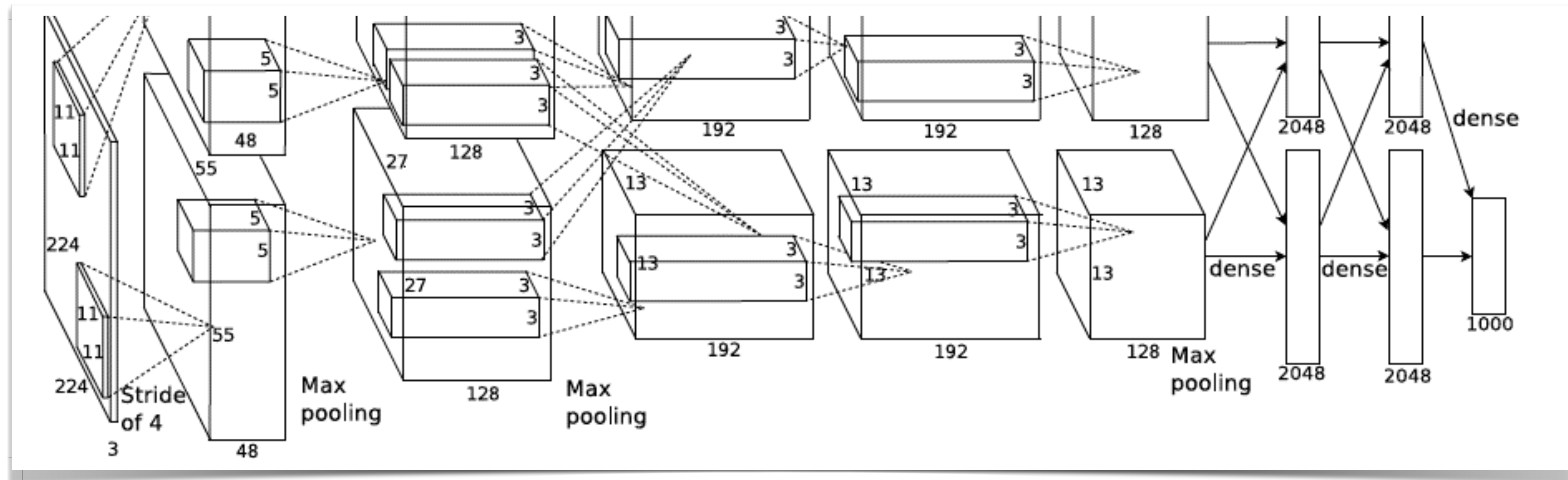
7 hidden layers, 650.000 units, 60 million parameter

More Data:

10^6 vs 10^3 images

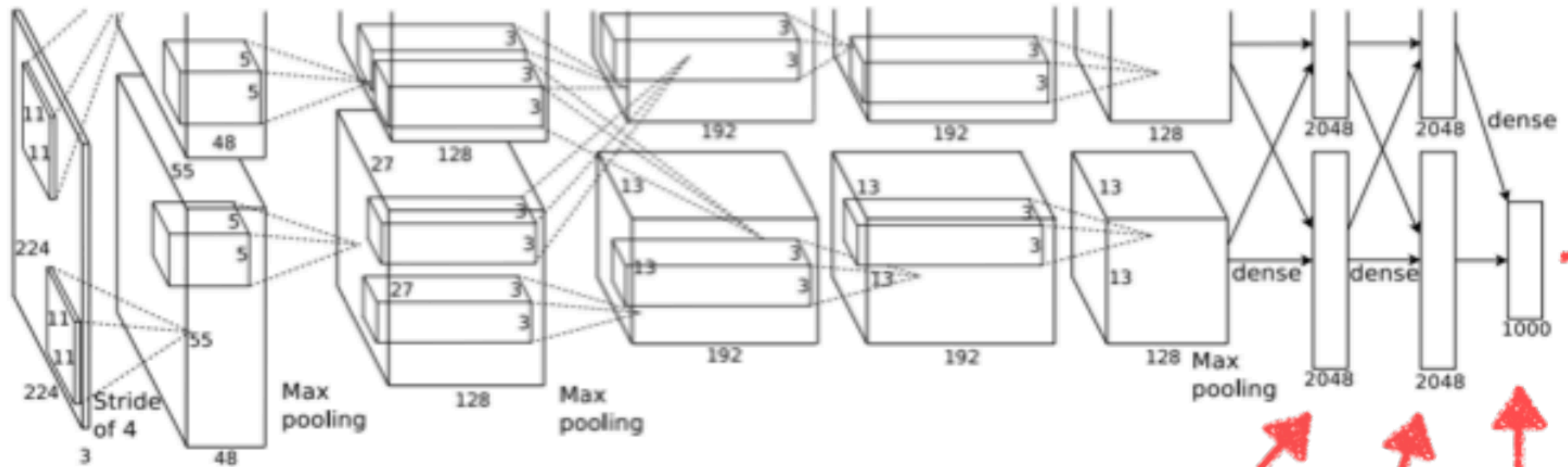
GPU implementation (50x speedup over CPU)

Similar framework to LeCun'98 but:



AlexNet

5 Convolutional Layers

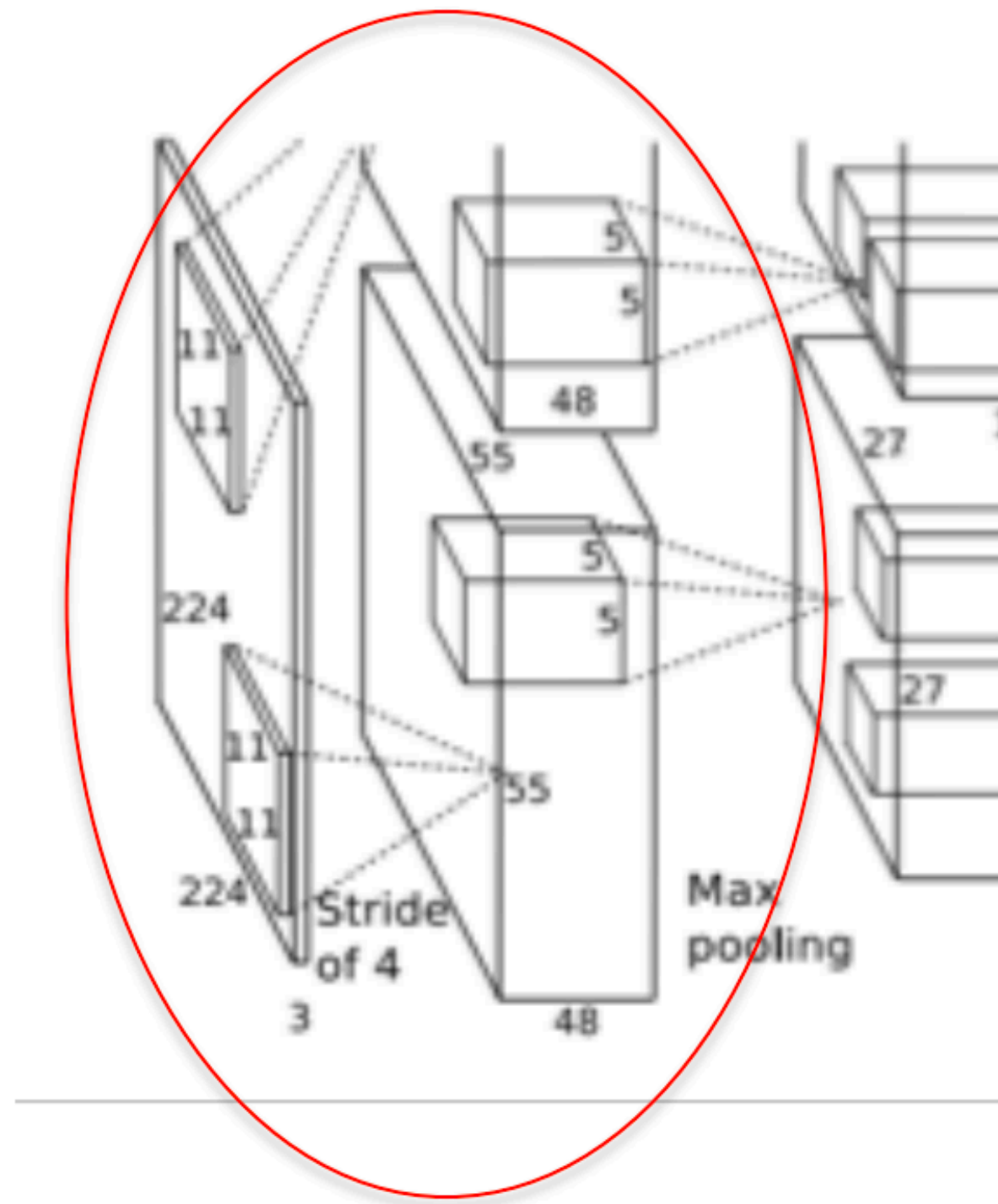


1000-way
softmax

3 Fully Connected Layers

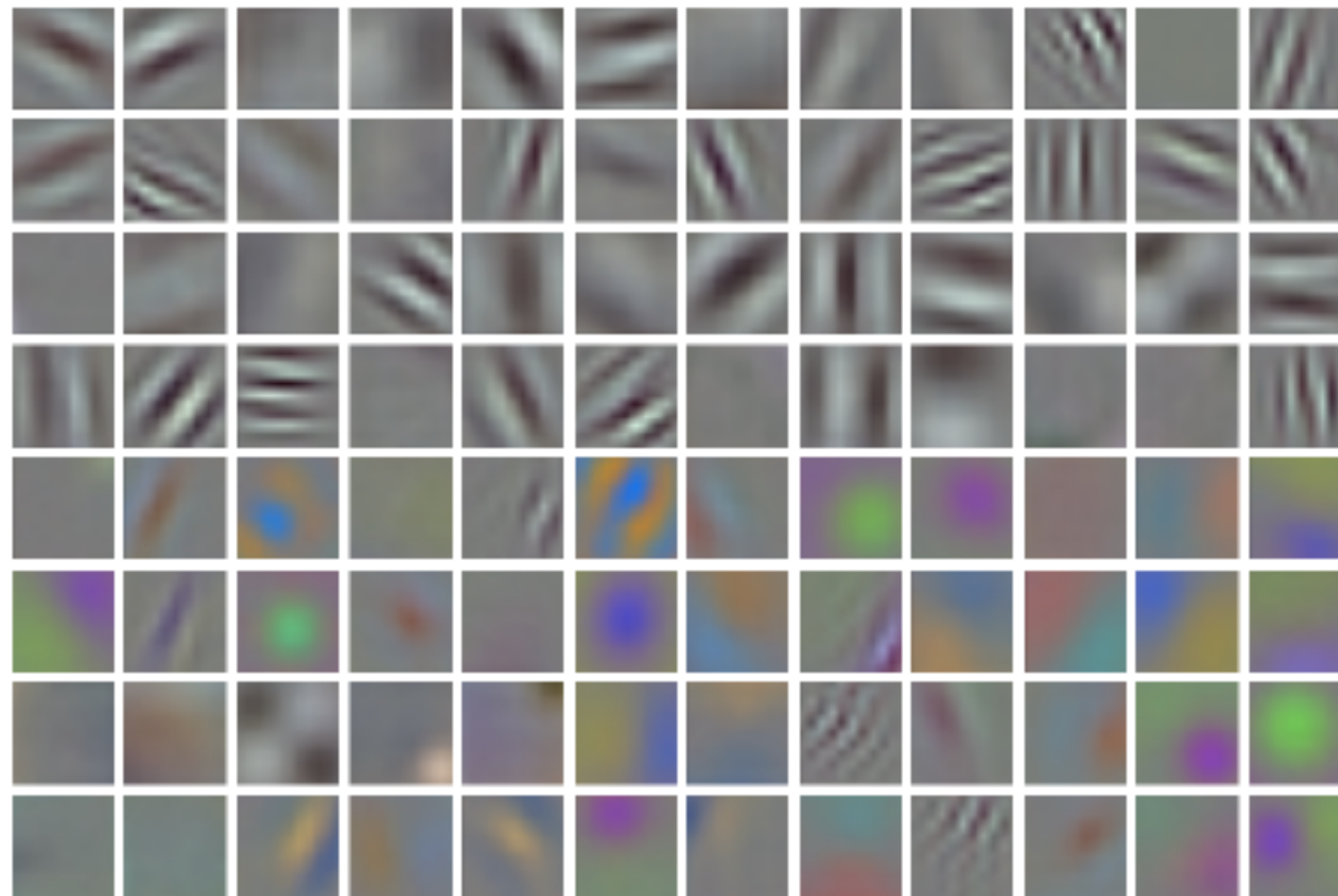
AlexNet

1st Convolution Layer

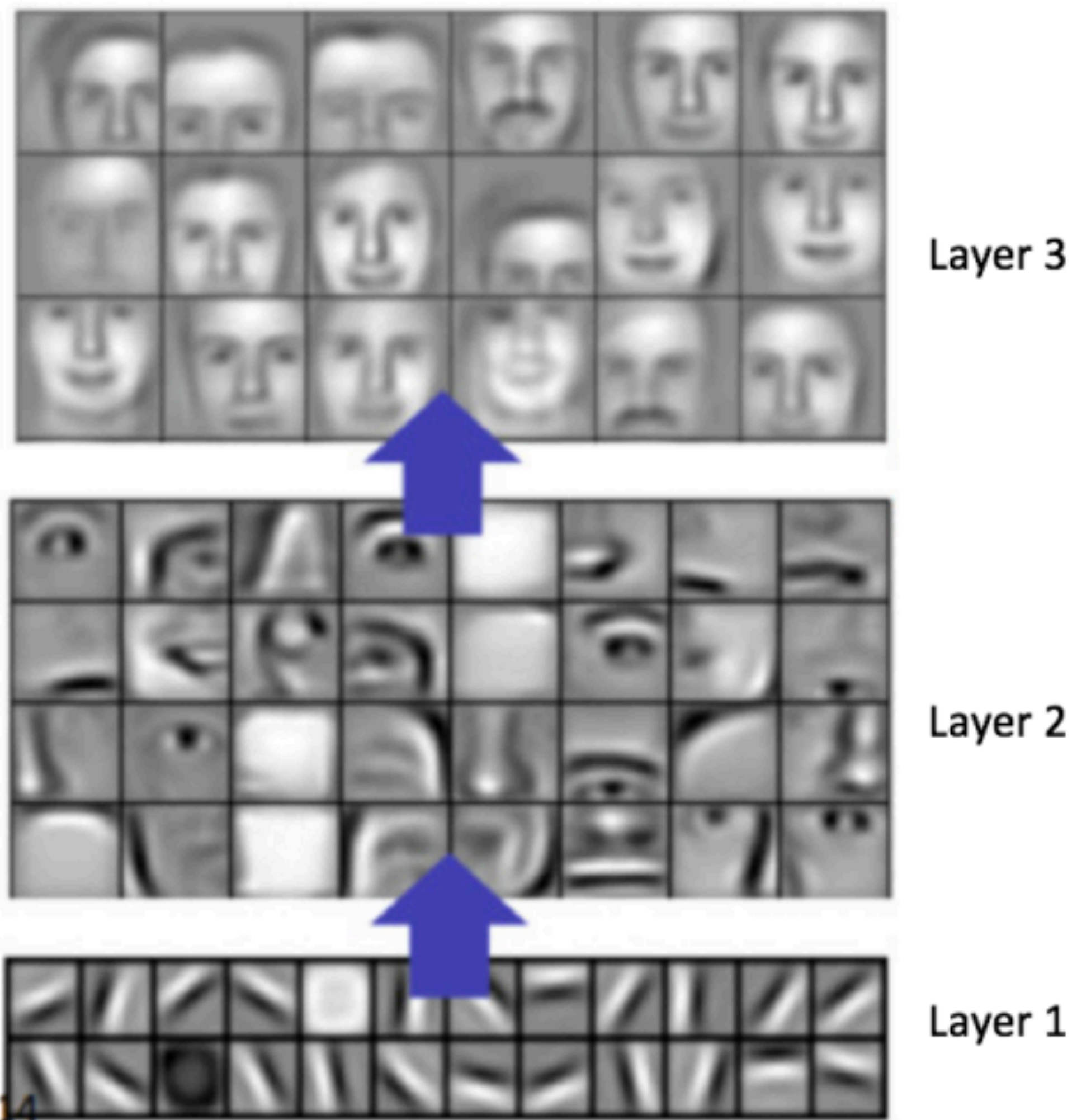


- Images: $227 \times 227 \times 3$
- F (receptive field size): 11
- S (stride) = 4
- Conv layer output: $55 \times 55 \times 96$

Alexnet 1st Conv Filters



Alexnet



Reducing Overfitting

Data Augmentation

60 million parameters, 650,000 neurons
-> overfits a lot

Crop 224x224 patches (and their horizontal reflections)

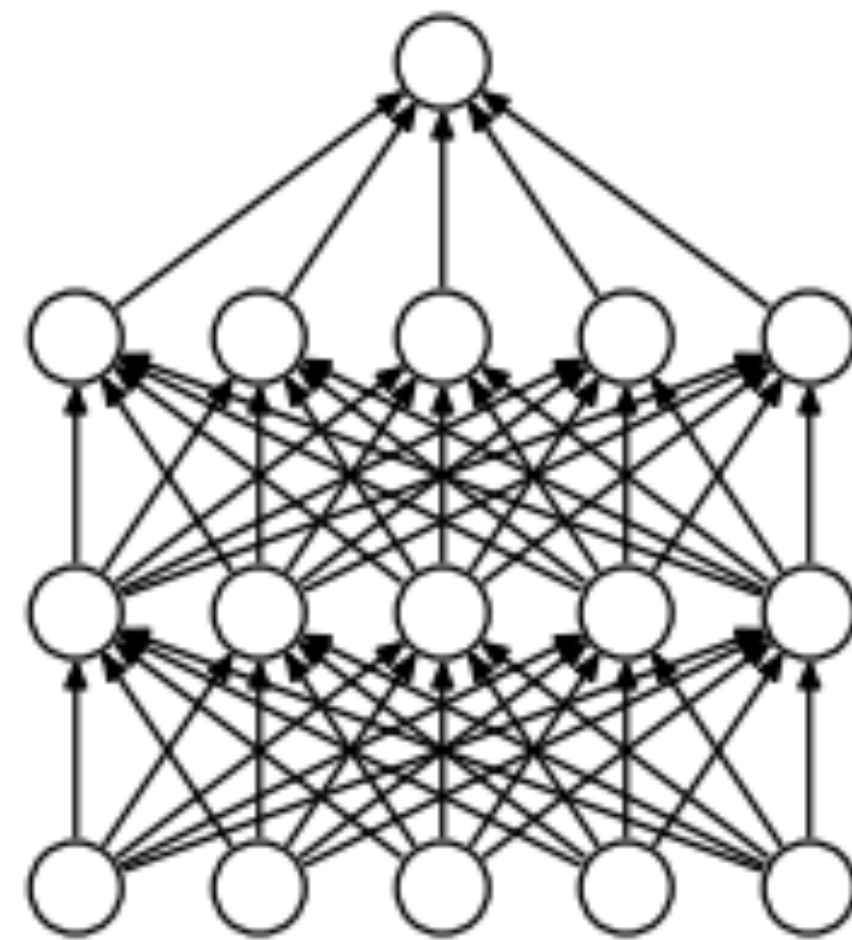
Data Augmentation at test

average the predictions on the 10 patches.

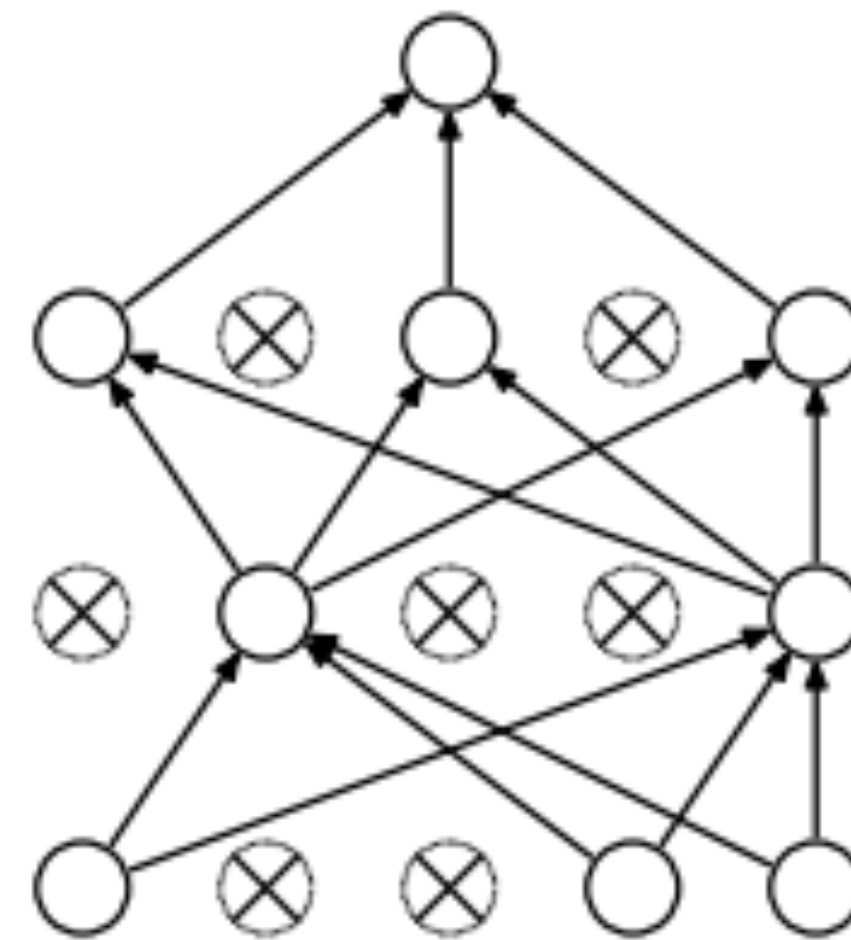
Reducing Overfitting

Dropout: A Simple Way to Prevent Neural Networks from Overfitting

Journal of Machine Learning Research 15 (2014) 1929-1958



(a) Standard Neural Net



(b) After applying dropout.

Exercise

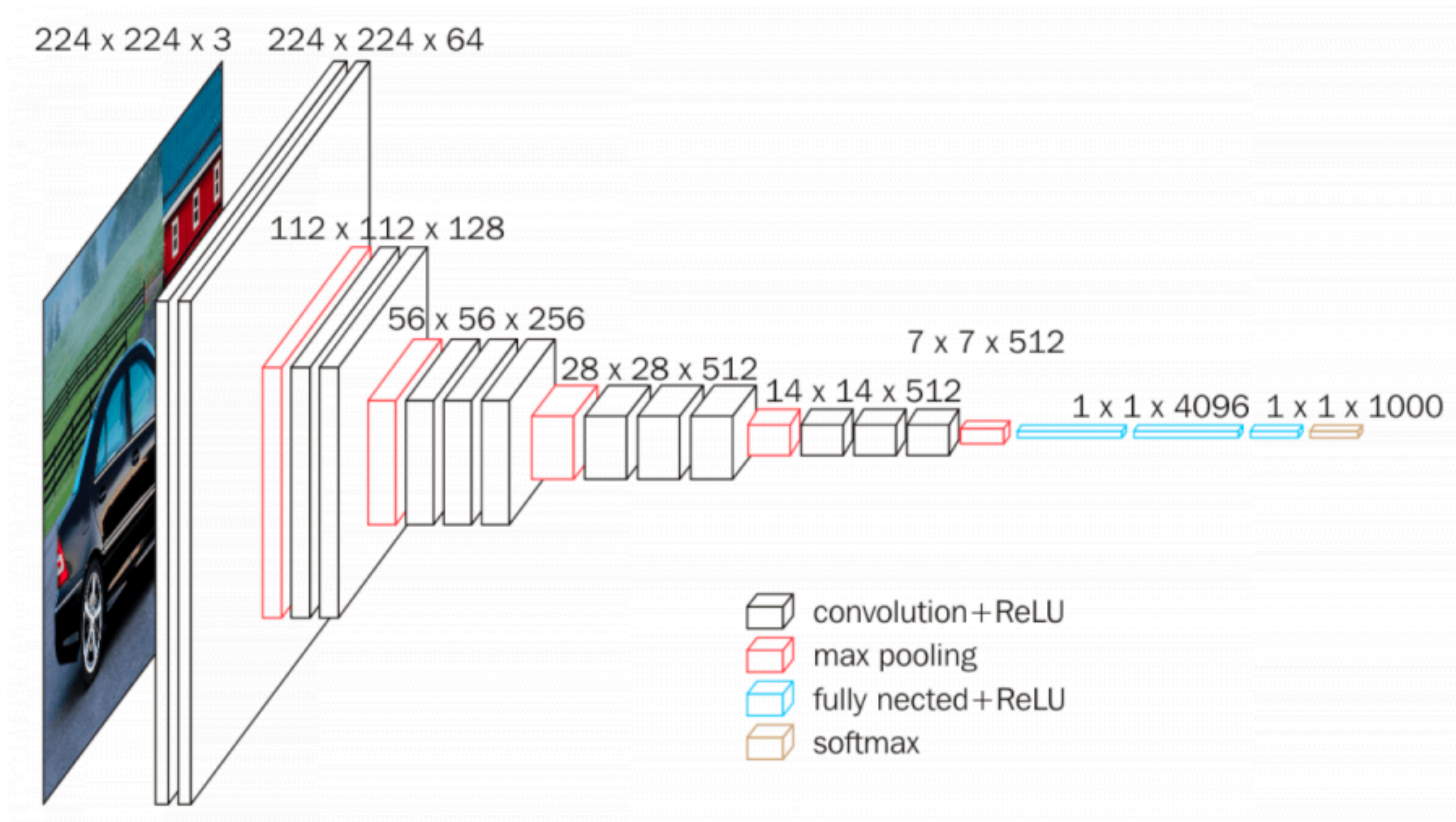
- How many parameter do the following model have if the input is a RGB image of 50x50:
 - 1 convolution layer with 10 filters of size 5x5
 - Max-pooling 2x2 with stride 2
 - 1 convolution layer with 20 filters of size 3x3
 - Dense Layer with 100 neurons
 - Dense Layer with 10 neurons

A meme featuring Leonardo DiCaprio from the movie Inception. He is shown from the chest up, wearing a dark suit, white shirt, and patterned tie. He has a serious, intense expression and is looking slightly to his right. The background is dark and out of focus, suggesting an indoor setting. The text "WE NEED TO GO" is overlaid in white, bold, sans-serif font at the top. The text "DEEPER" is overlaid in the same font at the bottom center. A small watermark "troll-face.ru" is visible in the bottom right corner.

WE NEED TO GO

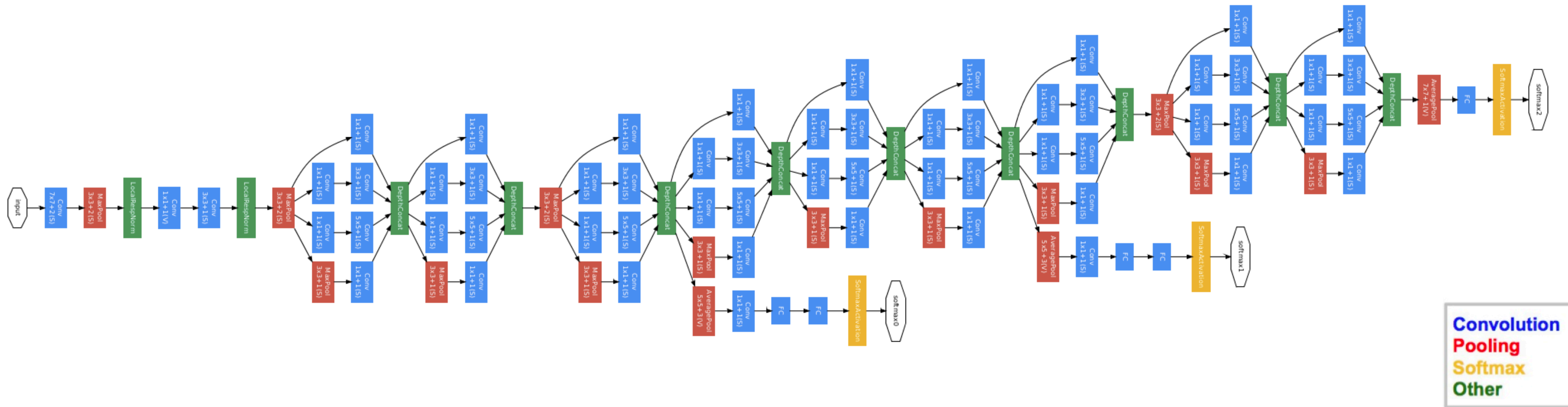
DEEPER

VGG Net



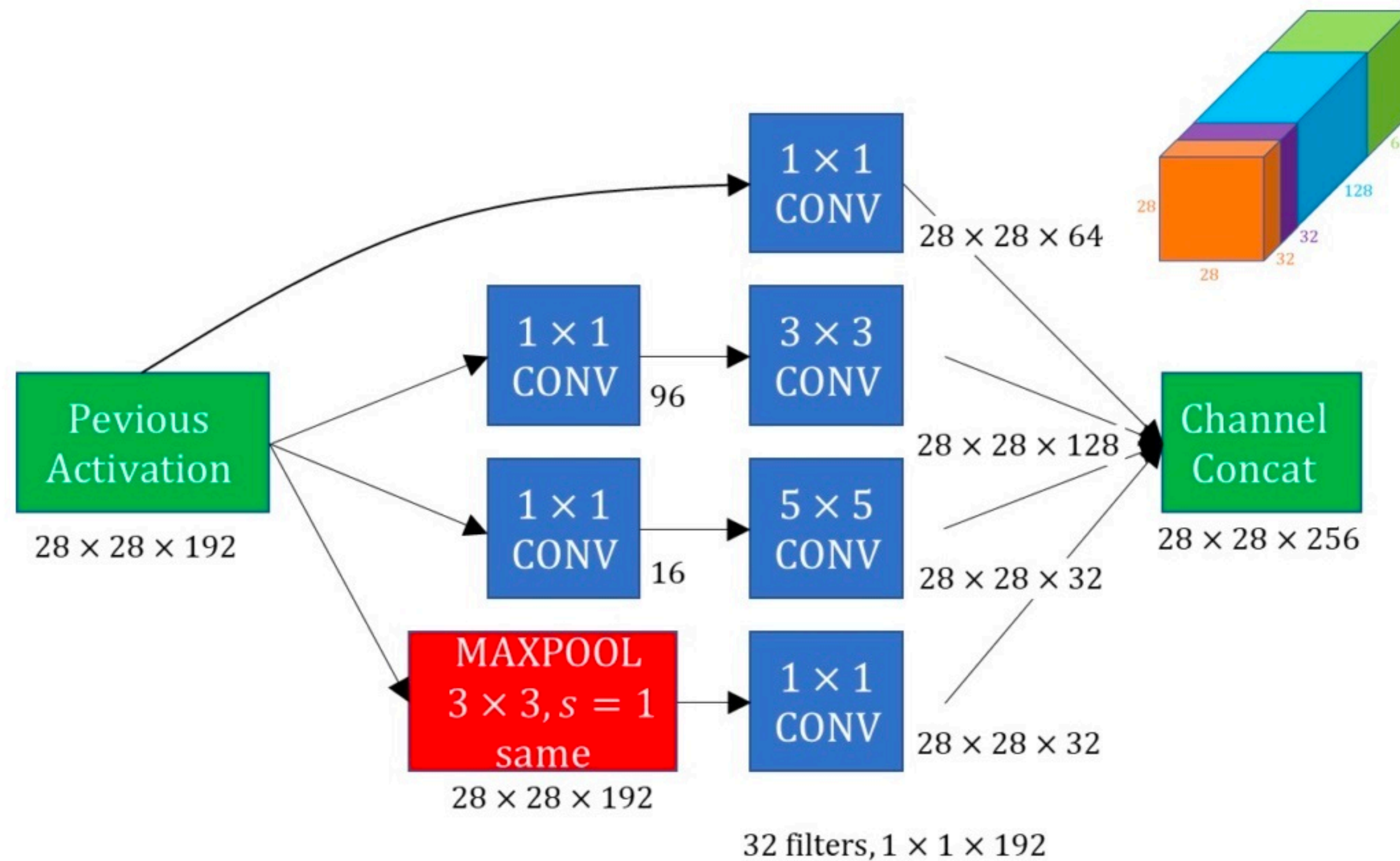
GoogleNet

22 layers, but **12 times less parameters** than AlexNet



GoogleNet

An example of an Inception module

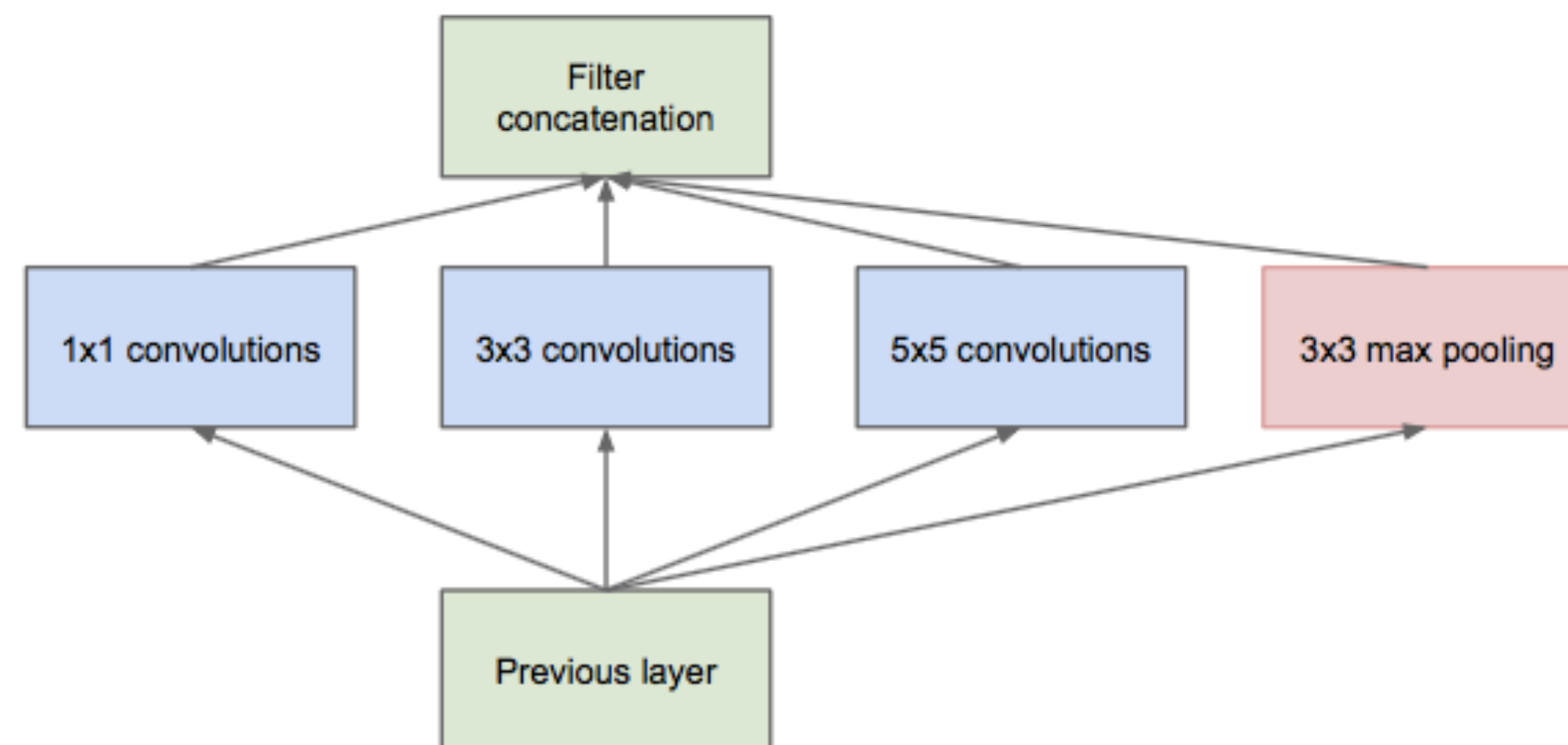


Inception Module: Naive Version

Convolutional filters with different sizes can cover different clusters of information.

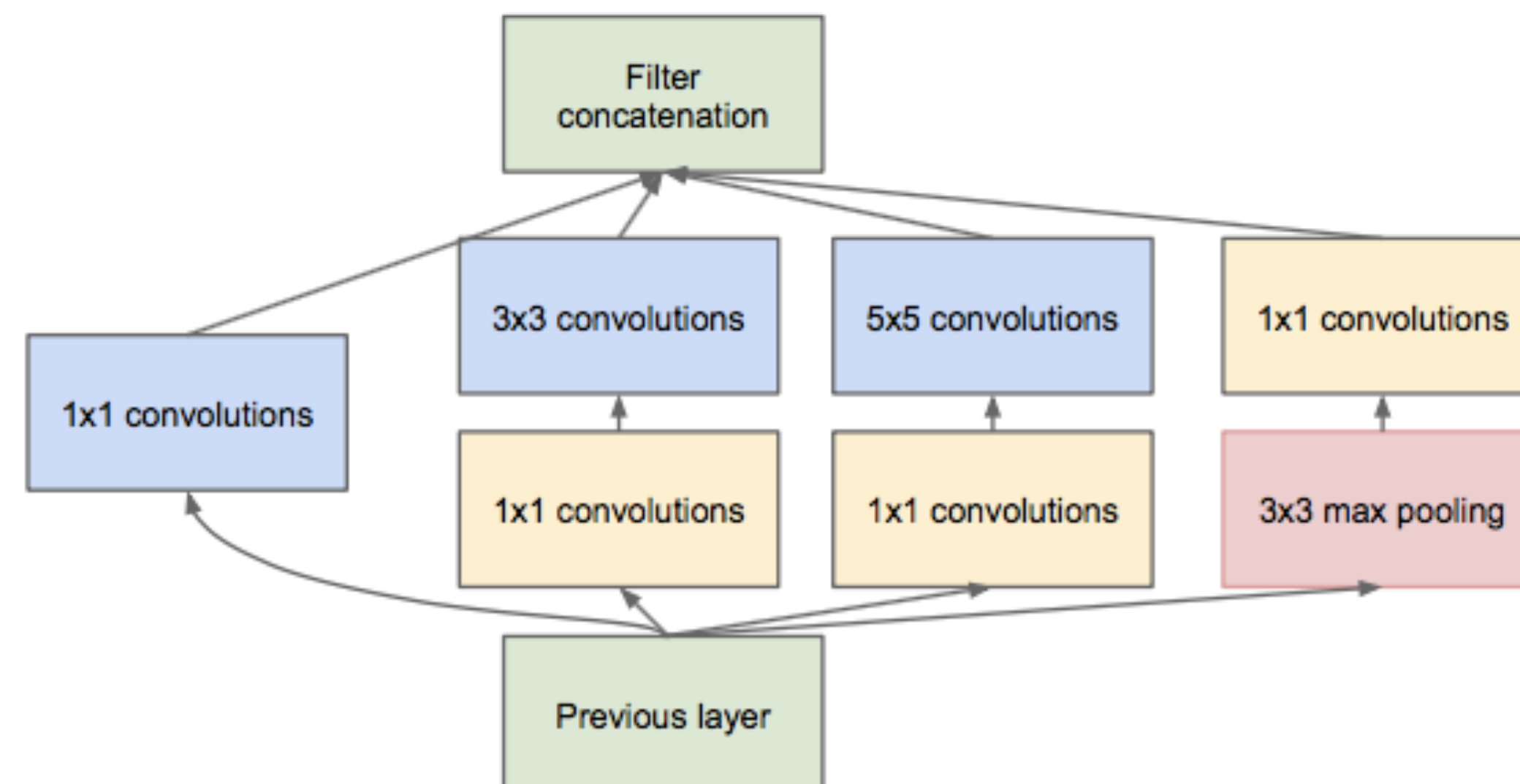
By finding the optimal local construction and repeating it spatially, they approximate the optimal sparse structure with dense components.

For convenience of computation, they use 1×1 , 3×3 and 5×5 filters + pooling.
Together these made up the naive Inception module.



Inception Module: Naive Version

Stacking these inception modules on top of each would lead to an exploding number of outputs
Solution: inspired by "Network in Network" add 1x1 convolutions for dimensionality reduction



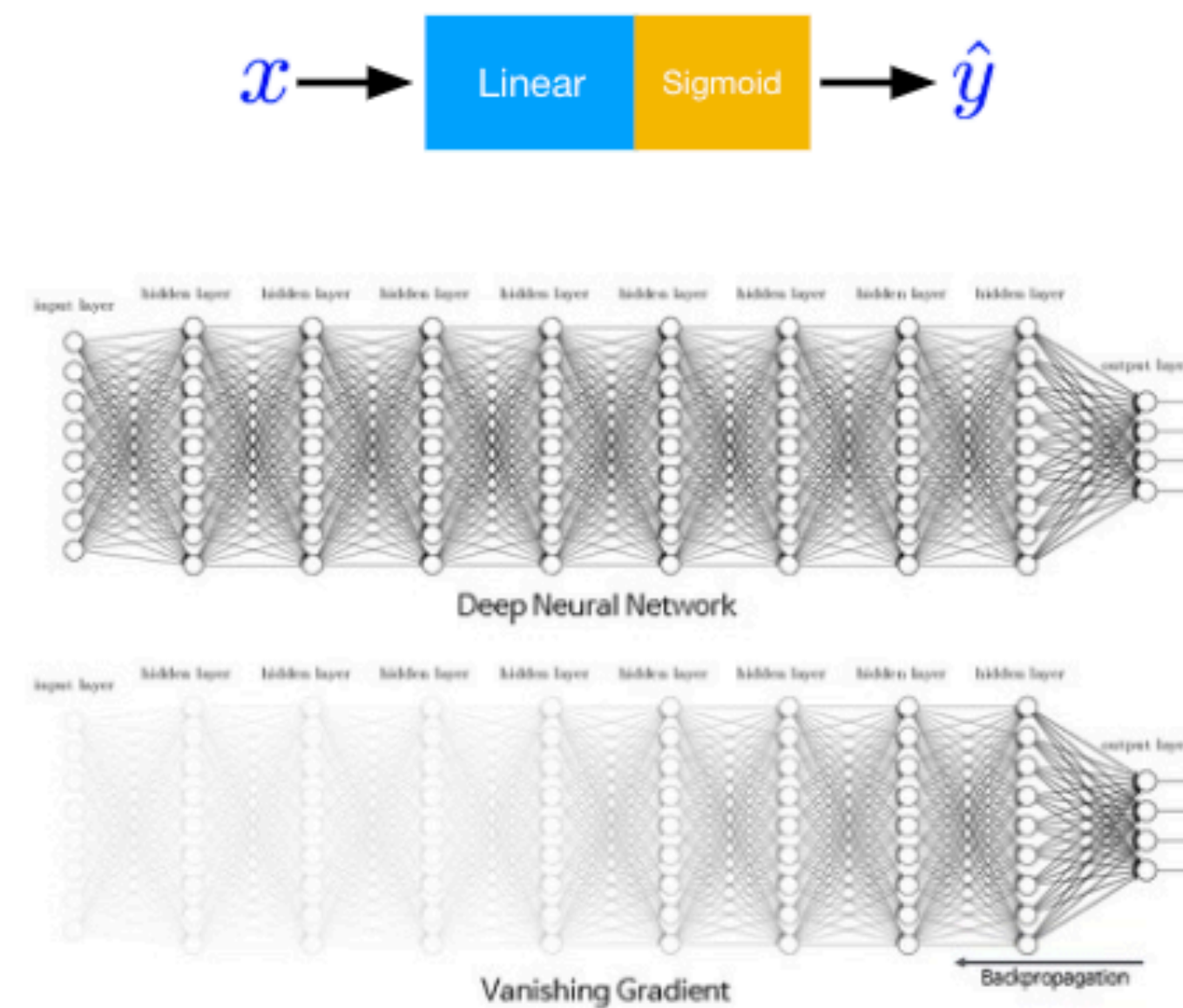
Conclusion:
More layers is better

A close-up shot of Leonardo DiCaprio in a dark suit, white shirt, and patterned tie. He has a serious, intense expression and is looking slightly to his right. The background is blurred, showing what appears to be an office or a formal setting with some architectural lines.

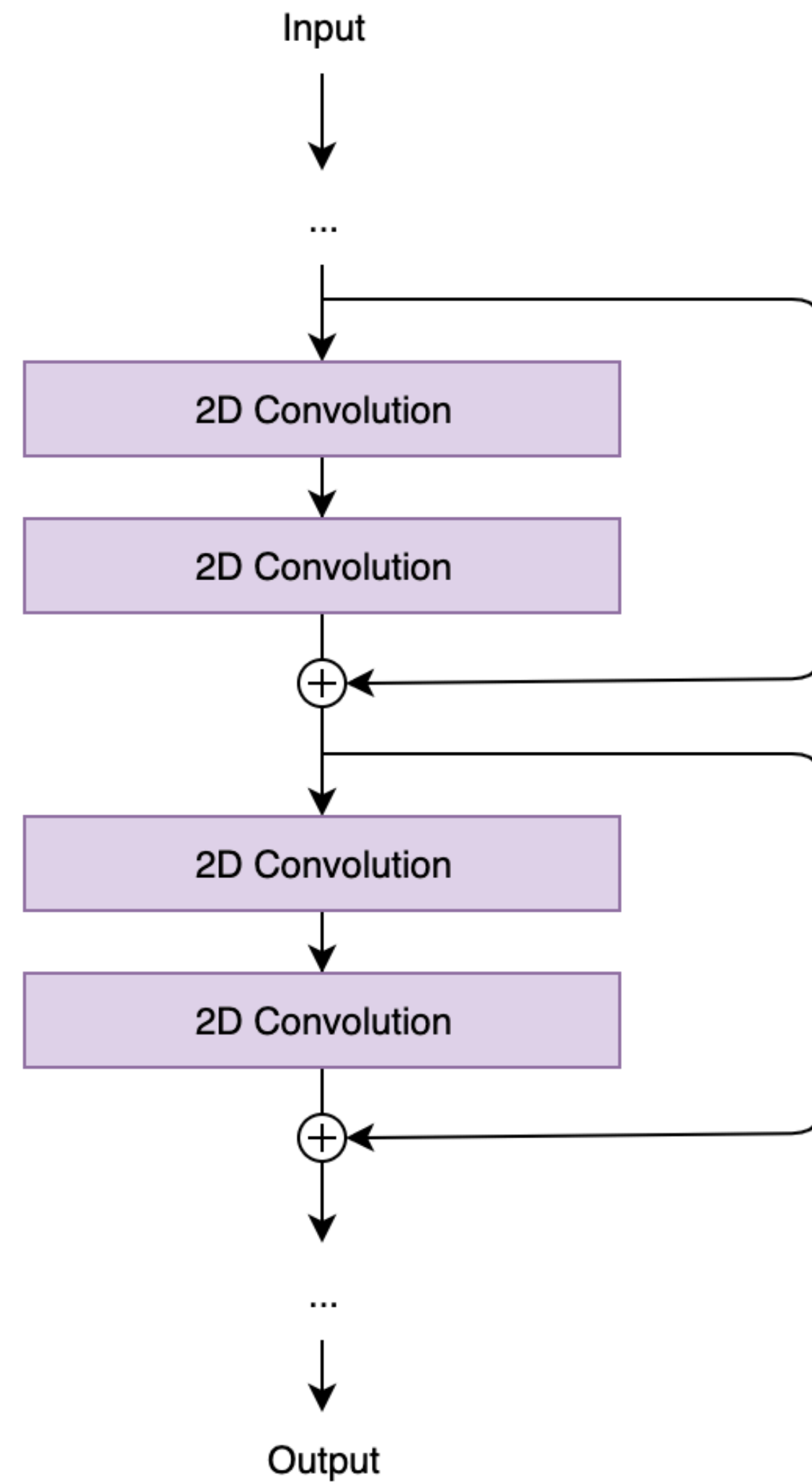
WE NEED TO GO

DEEPER

Not everything is that simple: **Vanishing Gradient Problem**

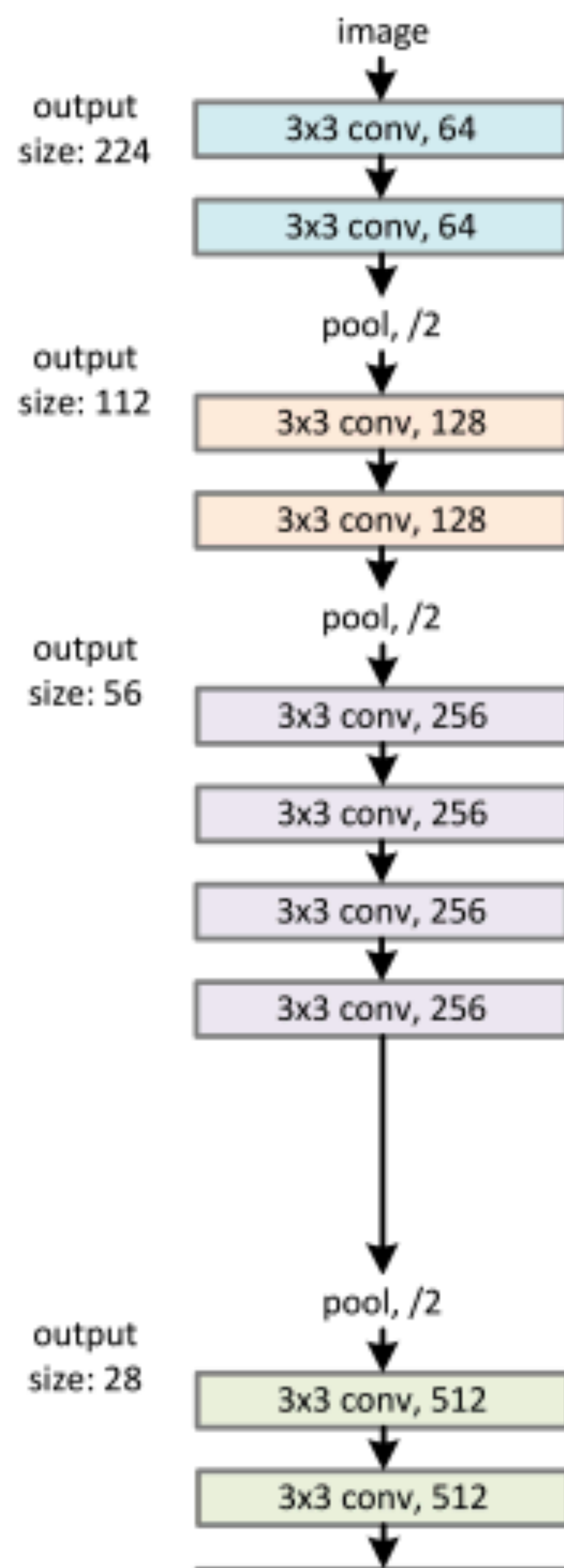


As the gradient keeps flowing backward to the initial layers, this value keeps getting multiplied by each local gradient. Hence, the gradient becomes smaller and smaller, making the updates to the initial layers very small, increasing the training time considerably.

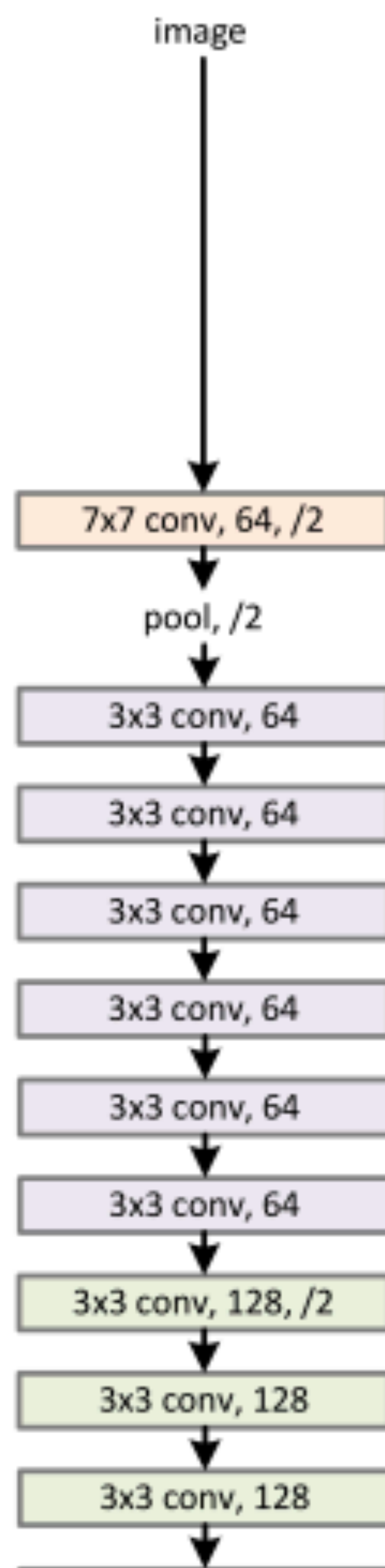


These ***skip connections*** act as gradient *superhighways*, allowing the gradient to flow unhindered.

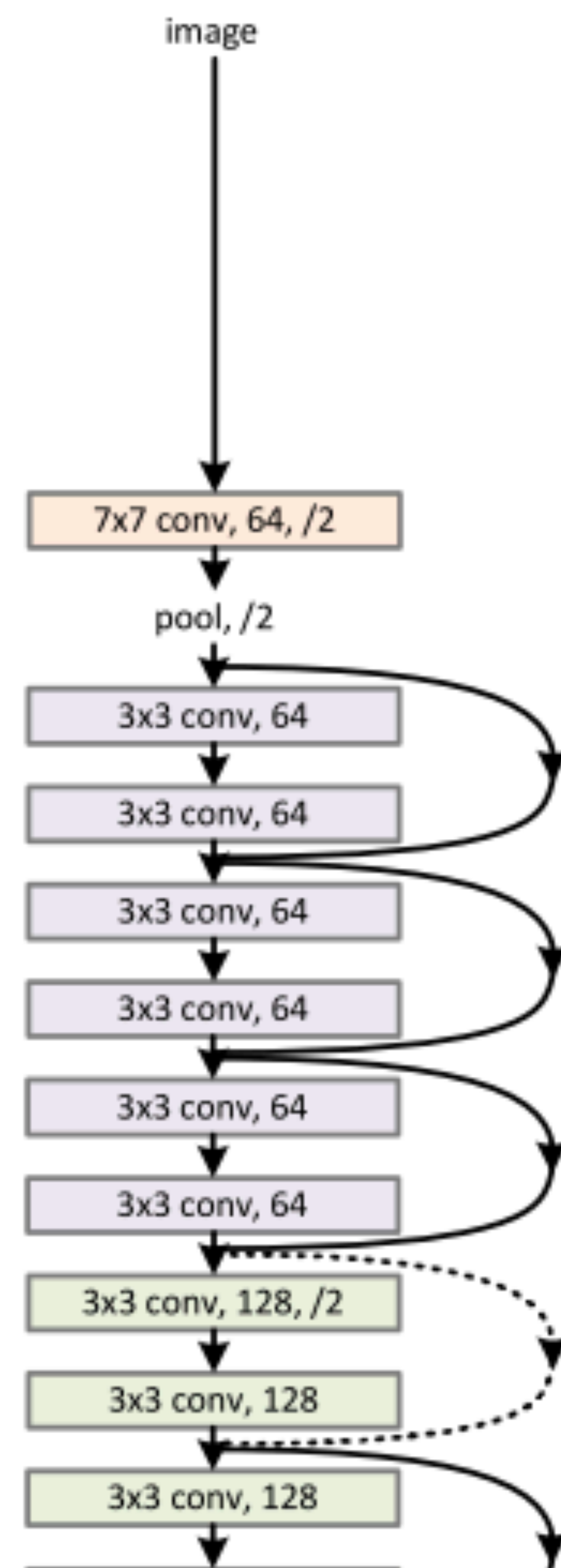
VGG-19



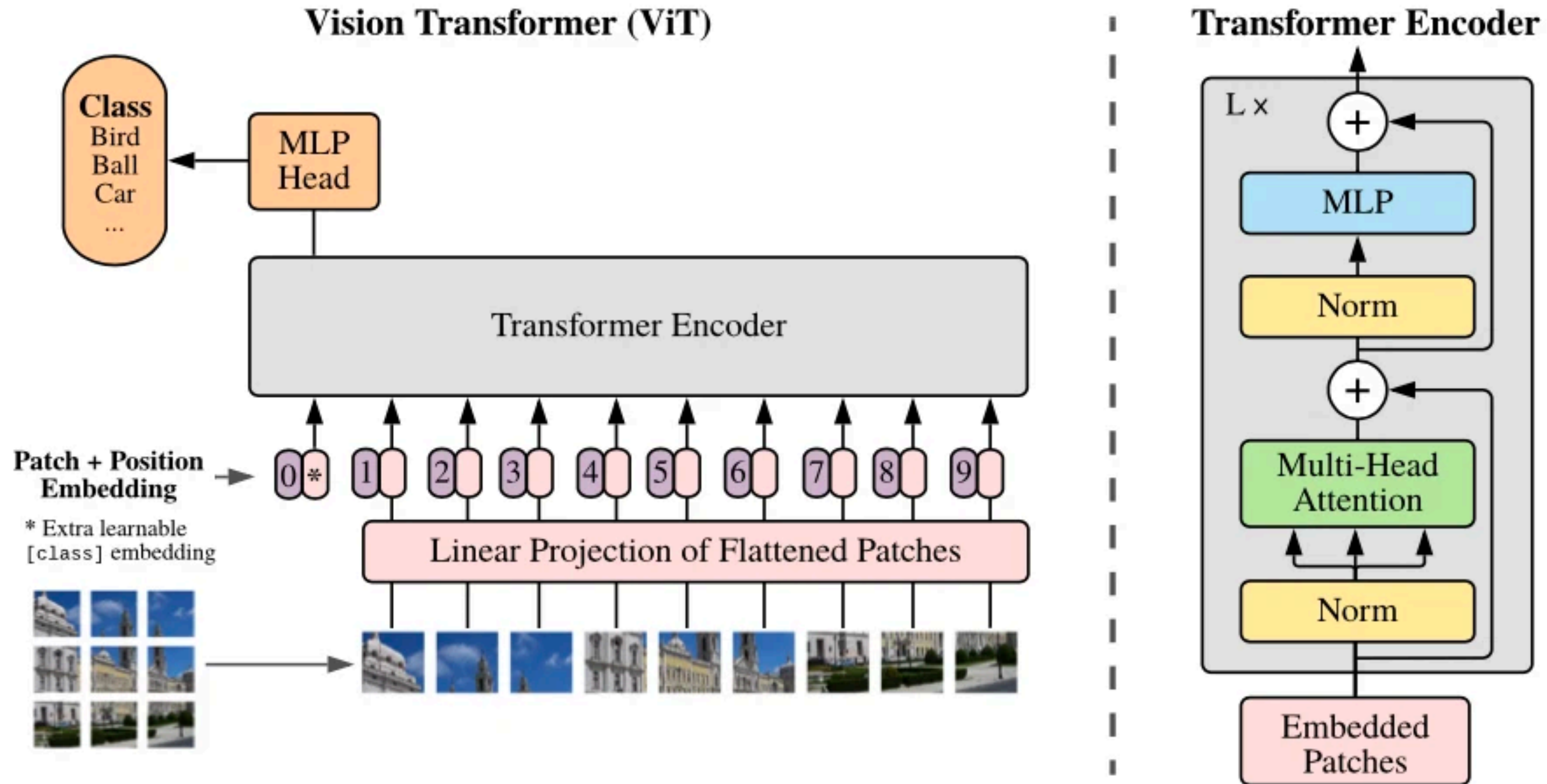
34-layer plain



34-layer residual



Visual TRANSFORMERS



Learning algorithm

- Learning the parameters is a complex and time-consuming task.
 - **Huge amount** of **data** is needed.
 - **Back progradation** method is used.
 - Great **packages** like Caffe, Theano or TensorFlow.
 - The make use of **GPU's** in order to optimize the weights.